ForEveryOne

File

Name :	Shaft2		
Changed by:	Matin	on: 26.01.2023	at: 01:17:47

Analysis of shafts, axle and beams

Input data

Coordinate system shaft: see picture W-002

Label		Shaft 1
Drawing		
Initial position (mm)		0.000
Length (mm)		450.000
Speed (1/min)		228.80
Sense of rotation: clockwise		
Material		C45 (1)
Young's modulus (N/mm²)		206000.000
Poisson's ratio nu		0.300
Density (kg/m³)		7830.000
Coefficient of thermal expansion	(10^-6/K)	11.500
Temperature (°C)		20.000
Weight of shaft (kg)		9.962
(Notice: Weight stands for the shaft only without considering the gears)		
Weight of shaft, including additional masses (kg)		34.728
Mass moment of inertia (kg*m²)		0.037
Momentum of mass GD2 (Nm²)		1.445
Position in space (°)		0.000
Gears mounted with stiffness according to ISO		
Consider deformations due to shearing		
Shear correction factor		1.100
Contact angle of rolling bearings is considered		
Tolerance field: Mean value		
Reference temperature (°C)		20.000

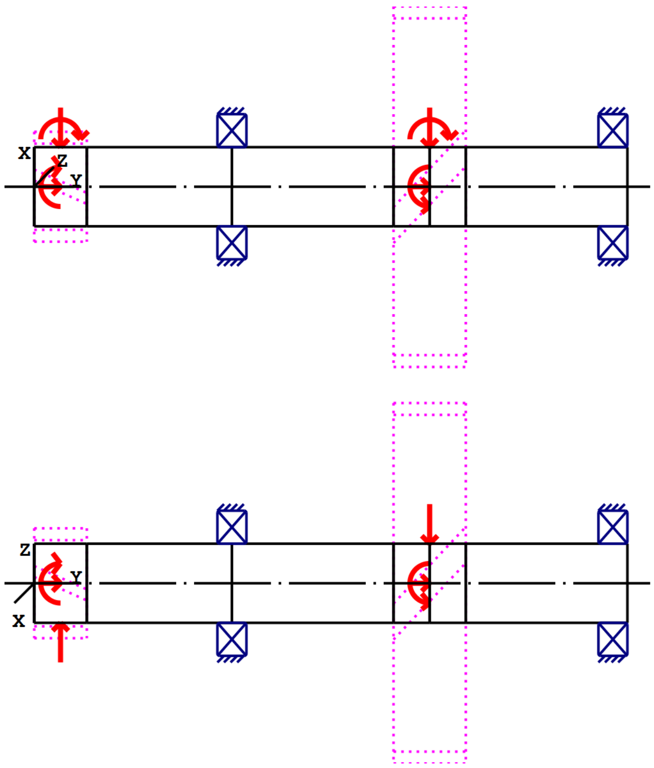


Figure: Load applications

Shaft definition (Shaft 1)

Outer contour

Cylinder (Cylinder)			0.000mm ... 150.000mm
Diameter (mm)	[d]	60.0000	
Length (mm)	[l]	150.0000	
Surface roughness (µm)	[Rz]	8.0000	

Cylinder (Cylinder)			150.000mm ... 300.000mm
Diameter (mm)	[d]	60.0000	
Length (mm)	[l]	150.0000	
Surface roughness (µm)	[Rz]	8.0000	

Cylinder (Cylinder)			300.000mm ... 450.000mm
Diameter (mm)	[d]	60.0000	
Length (mm)	[l]	150.0000	
Surface roughness (µm)	[Rz]	8.0000	

Forces

Type of force element		**Cylindrical gear**
Label in the model		Cylindrical gear
Position on shaft (mm)	[y_{local}]	300.0000
Position in global system (mm)	[y_{global}]	300.0000
Operating pitch diameter (mm)		273.2338
Helix angle (°)		25.1456 left
Working pressure angle at normal section (°)		20.8359

Position of contact (°)		0.0000
Length of load application (mm)		54.8000
Power (kW)		3.0000 driven (input)
Torque (Nm)		125.2093
Axial force (N)		430.2093
Shearing force X (N)		-385.3198
Shearing force Z (N)		-916.4993
Bending moment X (Nm)		-0.0000
Bending moment Z (Nm)		58.7739

Type of force element		**Cylindrical gear**
Label in the model		Cylindrical gear
Position on shaft (mm)	[y_{local}]	20.0000
Position in global system (mm)	[y_{global}]	20.0000
Operating pitch diameter (mm)		83.3030
Helix angle (°)		25.1456 right
Working pressure angle at normal section (°)		20.8359
Position of contact (°)		0.0000
Length of load application (mm)		39.8780
Power (kW)		3.0000 driving (output)
Torque (Nm)		-125.2093
Axial force (N)		1411.0864
Shearing force X (N)		-1263.8490
Shearing force Z (N)		3006.1178
Bending moment X (Nm)		-0.0000
Bending moment Z (Nm)		58.7739

Bearing

Label in the model		Rolling bearing
Bearing type		SKF 6212
Bearing type		Deep groove ball bearing (single row)
		SKF EXPLORER
Bearing position (mm)	[y_{lokal}]	439.000
Bearing position (mm)	[y_{global}]	439.000
Attachment of external ring		Set fixed bearing right
Inner diameter (mm)	[d]	60.000
External diameter (mm)	[D]	110.000
Width (mm)	[b]	22.000
Corner radius (mm)	[r]	1.500
Basic static load rating (kN)	[C_0]	36.000
Basic dynamic load rating (kN)	[C]	55.300
Fatigue load rating (kN)	[C_u]	1.500
Values for approximated geometry:		
Basic dynamic load rating (kN)	[C_{theo}]	0.000
Basic static load rating (kN)	[C_{0theo}]	0.000
Correction factor Basic dynamic load rating	[f_C]	1.000
Correction factor Basic static load rating	[f_{C0}]	1.000

Label in the model		Rolling bearing
Bearing type		SKF 6212
Bearing type		Deep groove ball bearing (single row)

		SKF EXPLORER
Bearing position (mm)	[y_{lokal}]	150.000
Bearing position (mm)	[y_{global}]	150.000
Attachment of external ring		Set fixed bearing left
Inner diameter (mm)	[d]	60.000
External diameter (mm)	[D]	110.000
Width (mm)	[b]	22.000
Corner radius (mm)	[r]	1.500
Basic static load rating (kN)	[C_0]	36.000
Basic dynamic load rating (kN)	[C]	55.300
Fatigue load rating (kN)	[C_u]	1.500
Values for approximated geometry:		
Basic dynamic load rating (kN)	[C_{theo}]	0.000
Basic static load rating (kN)	[C_{0theo}]	0.000
Correction factor Basic dynamic load rating	[f_C]	1.000
Correction factor Basic static load rating	[f_{C0}]	1.000

Shaft 'Shaft 1': Cylindrical gear 'Cylindrical gear' (y= 286.3000 (mm)) is taken into account as component of the shaft.
EI (y= 272.6000 (mm)): 131051.5375 (Nm²), EI (y= 300.0000 (mm)): 131051.5375 (Nm²), m (yS= 286.3000 (mm)): 11.9731 (kg)
Jp: 0.0160 (kg*m²), Jxx: 0.0082 (kg*m²), Jzz: 0.0082 (kg*m²)

Shaft 'Shaft 1': Cylindrical gear 'Cylindrical gear' (y= 313.7000 (mm)) is taken into account as component of the shaft.
EI (y= 300.0000 (mm)): 131051.5375 (Nm²), EI (y= 327.4000 (mm)): 131051.5375 (Nm²), m (yS= 313.7000 (mm)): 11.9731 (kg)
Jp: 0.0160 (kg*m²), Jxx: 0.0082 (kg*m²), Jzz: 0.0082 (kg*m²)

Shaft 'Shaft 1': Cylindrical gear 'Cylindrical gear' (y= 20.0000 (mm)) is taken into account as component of the shaft.
EI (y= 0.0610 (mm)): 131051.5375 (Nm²), EI (y= 39.9390 (mm)): 131051.5375 (Nm²), m (yS= 20.0000 (mm)): 0.8189 (kg)
Jp: 0.0004 (kg*m²), Jxx: 0.0003 (kg*m²), Jzz: 0.0003 (kg*m²)

Results

Shaft

Maximum deflection (μm)	83.937
Position of the maximum (mm)	0.000
Mass center of gravity (mm)	225.000
Total axial load (N)	1841.296
Torsion under torque (°)	0.017

Bearing

Probability of failure	[n]	10.00	%
Axial clearance	[u_A]	10.00	μm

Lubricant	Oil: ISO-VG 220		
Lubricant - service temperature	[T_B]	20.00	°C

Rolling bearings, classical calculation (contact angle considered)

Shaft 'Shaft 1' Rolling bearing 'Rolling bearing'

Position (Y-coordinate)	[y]	439.00	mm
Dynamic equivalent load	[P]	4.16	kN
Equivalent load	[P_0]	2.06	kN
Life modification factor for reliability[a_1]		1.000	
Basic bearing rating life	[L_{nh}]	170966.60	h
Operating viscosity	[ν]	912.87	mm²/s
Static safety factor	[S_0]	17.51	
Bearing reaction force	[Fx]	0.038	kN
Bearing reaction force	[Fy]	-1.841	kN
Bearing reaction force	[Fz]	1.893	kN
Bearing reaction force	[Fr]	1.893	kN (88.84°)
Bearing reaction moment	[Mx]	0.00	Nm
Bearing reaction moment	[My]	0.00	Nm
Bearing reaction moment	[Mz]	0.00	Nm
Bearing reaction moment	[Mr]	0.00	Nm (20.56°)
Oil level	[H]	0.000	mm
Rolling moment of friction	[M_{rr}]	0.559	Nm
Sliding moment of friction	[M_{sl}]	0.099	Nm
Moment of friction, seals	[M_{seal}]	0.000	Nm

Moment of friction for seals determined according to SKF main catalog 17000/1 EN:2018

Moment of friction flow losses	[M_{drag}]	0.000	Nm
Torque of friction	[M_{loss}]	0.658	Nm
Power loss	[P_{loss}]	15.759	W

The moment of friction is calculated according to the details in SKF Catalog 2018.

The calculation is always performed with a coefficient for additives in the lubricant μbl=0.15.

Displacement of bearing	[u_x]	-0.272	μm
Displacement of bearing	[u_y]	10.000	μm
Displacement of bearing	[u_z]	-8.996	μm
Displacement of bearing	[u_r]	9.000	μm (-91.73°)
Misalignment of bearing	[r_x]	0.065	mrad (0.22')
Misalignment of bearing	[r_y]	0.302	mrad (1.04')
Misalignment of bearing	[r_z]	0.002	mrad (0.01')
Misalignment of bearing	[r_r]	0.065	mrad (0.22')

Shaft 'Shaft 1' Rolling bearing 'Rolling bearing'

Position (Y-coordinate)	[y]	150.00	mm
Dynamic equivalent load	[P]	4.13	kN
Equivalent load	[P_0]	4.13	kN
Life modification factor for reliability[a_1]		1.000	
Basic bearing rating life	[L_{nh}]	175032.63	h
Operating viscosity	[ν]	912.87	mm²/s
Static safety factor	[S_0]	8.72	
Bearing reaction force	[Fx]	1.611	kN
Bearing reaction force	[Fy]	0.000	kN
Bearing reaction force	[Fz]	-3.802	kN
Bearing reaction force	[Fr]	4.129	kN (-67.04°)
Bearing reaction moment	[Mx]	-0.00	Nm
Bearing reaction moment	[My]	0.00	Nm
Bearing reaction moment	[Mz]	0.00	Nm
Bearing reaction moment	[Mr]	0.00	Nm (90°)

Oil level	[H]	0.000	mm
Rolling moment of friction	[M_{rr}]	0.261	Nm
Sliding moment of friction	[M_{sl}]	0.054	Nm
Moment of friction, seals	[M_{seal}]	0.000	Nm

Moment of friction for seals determined according to SKF main catalog 17000/1 EN:2018

Moment of friction flow losses	[M_{drag}]	0.000	Nm
Torque of friction	[M_{loss}]	0.315	Nm
Power loss	[P_{loss}]	7.551	W

The moment of friction is calculated according to the details in SKF Catalog 2018.

The calculation is always performed with a coefficient for additives in the lubricant µbl=0.15.

Displacement of bearing	[u_x]	-3.461	µm
Displacement of bearing	[u_y]	10.670	µm
Displacement of bearing	[u_z]	8.308	µm
Displacement of bearing	[u_r]	9.000	µm (112.61°)
Misalignment of bearing	[r_x]	-0.341	mrad (-1.17')
Misalignment of bearing	[r_y]	0.149	mrad (0.51')
Misalignment of bearing	[r_z]	-0.075	mrad (-0.26')
Misalignment of bearing	[r_r]	0.349	mrad (1.2')

Damage (%) [Lreq] (31000.000)

Bin no	B1	B2
1	18.13	17.71

Σ	18.13	17.71

Utilization (%) [Lreq] (31000.000)

B1	B2
56.60	56.16

Note: Utilization = (Lreq/Lh)^(1/k)

Ball bearing: k = 3, roller bearing: k = 10/3

B1: Rolling bearing

B2: Rolling bearing

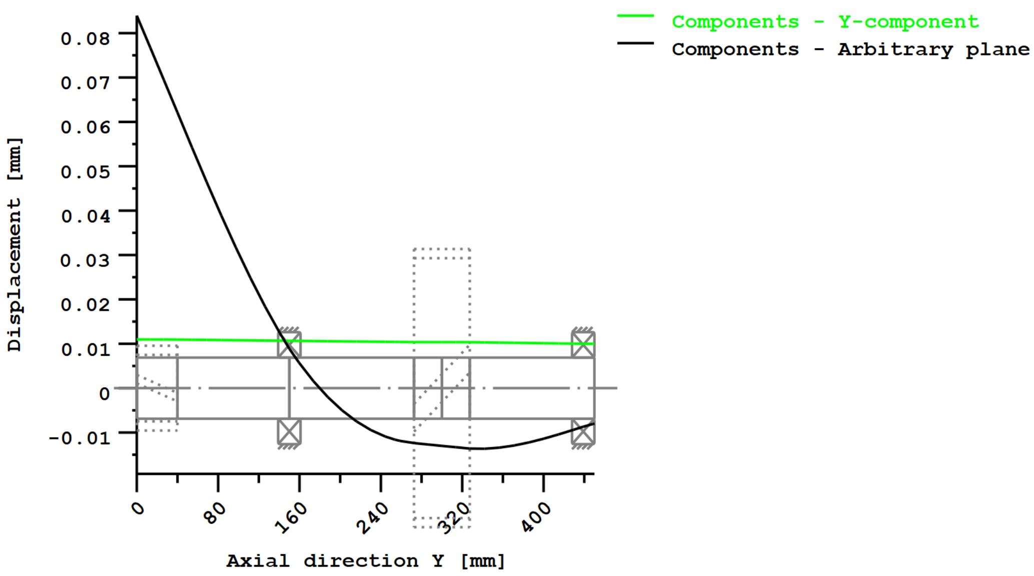
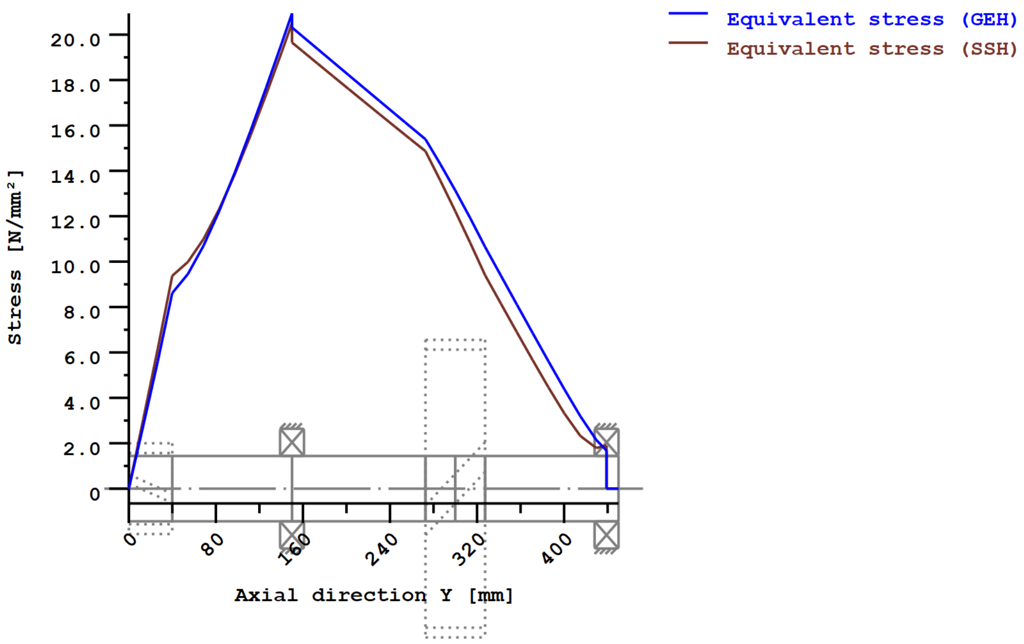


Figure: Deformation (bending etc.) (Arbitrary plane 103.3059018 121)



Nominal stresses, without taking into account stress concentrations
GEH(von Mises): sigV = ((sigB+sigZ,D)^2 + 3*(tauT+tauS)^2)^1/2
SSH(Tresca): sigV = ((sigB-sigZ,D)^2 + 4*(tauT+tauS)^2)^1/2

Figure: Equivalent stress

KISSsoft
Drivetrain Design Solutions

Strength calculation according to DIN 743:2012

Summary

Shaft 1

Material	C45 (1)
Material type	Through hardened steel
Material treatment	unalloyed, through hardened
Surface treatment	No

Calculation of endurance limit and the static strength

Calculation for load case 2 (σav/σmv = const)

Cross section	Position (Y-Coord) (mm)	
A-A	150.00	Smooth shaft

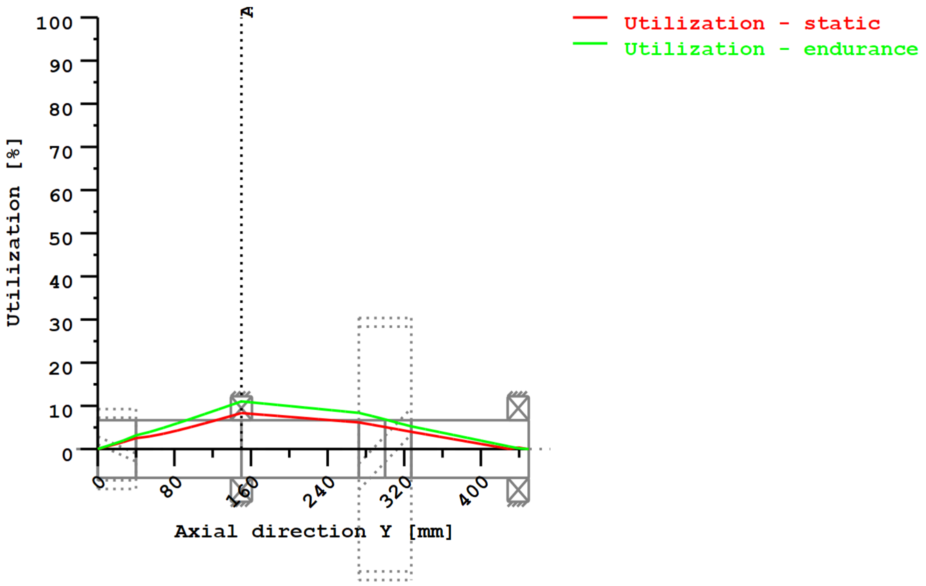
Results:

Cross section	Kfb	Kfσ	K2d	SD	SS
A-A	1.00	0.91	0.86	10.88	14.40
Required safeties:				1.20	1.20

Abbreviations:
Kfb: Notch factor bending
Kfσ: Surface factor
K2d: size factor bending
SD: Safety endurance limit
SS: Safety against yield point

Utilization (%) [Smin/S]

Cross section	Static	Endurance
A-A	8.336	11.032
Maximum utilization (%)	[A]	11.032



Utilization = Smin/S (%)

Figure: Strength

Calculation details

General statements

Label	Shaft 1		
Drawing			
Length (mm)		[l]	450.00
Speed (1/min)		[n]	228.80

Material	C45 (1)
Material type	Through hardened steel
Material treatment	unalloyed, through hardened
Surface treatment	No

	Tension/Compression	Bending	Torsion	Shearing
Load factor static calculation	1.700	1.700	1.700	1.700
Load factor endurance limit	1.000	1.000	1.000	1.000

Reference diameter material (mm)	[dB]	16.00
σB according to DIN 743 (at dB) (N/mm²)	[σB]	700.00
σS according to DIN 743 (at dB) (N/mm²)	[σS]	490.00
[σzdW] (bei dB) (N/mm²)		280.00
[σbW] (bei dB) (N/mm²)		350.00
[τtW] (bei dB) (N/mm²)		210.00
Thickness of raw material (mm)	[dWerkst]	65.00

Material data calculated according DIN743/3 with K1(d)
Material strength calculated from size of raw material
Geometric size factor K1d calculated from raw material diameter

[σBeff] (N/mm²)	589.20
[σSeff] (N/mm²)	388.58
[σbF] (N/mm²)	466.29
[τtF] (N/mm²)	269.21
[σBRand] (N/mm²)	628.00
[σzdW] (N/mm²)	235.68
[σbW] (N/mm²)	294.60
[τtW] (N/mm²)	176.76

Endurance limit for single stage use

Calculation for load case 2 (σ.av/σ.mv = const)

Cross section 'A-A' Smooth shaft

Comment		
Position (Y-Coordinate) (mm)	[y]	150.000
External diameter (mm)	[da]	60.000
Inner diameter (mm)	[di]	0.000
Notch effect		Smooth shaft
Mean roughness (µm)	[Rz]	8.000

		Tension/Compression	Bending	Torsion	Shearing
Load: (N) (Nm)					
Mean value	[Fzdm, Mbm, Tm, Fqm]	-705.4	0.0	62.6	0.0
Amplitude	[Fzda, Mba, Ta, Fqa]	705.4	401.9	62.6	3227.8
Maximum value	[Fzdmax, Mbmax, Tmax, Fqmax]	-2398.4	683.3	212.9	5487.3
Cross section, moment of resistance: (mm²)					
	[A, Wb, Wt, A]	2827.4	21205.8	42411.5	2827.4
Stresses: (N/mm²)					
	[σzdm, σbm, τm, τqm] (N/mm²)	-0.249	0.000	1.476	0.000
	[σzda, σba, τa, τqa] (N/mm²)	0.249	18.953	1.476	1.522
	[σzdmax,σbmax,τmax,τqmax] (N/mm²)	-0.848	32.220	5.019	2.588

Technological size influence	[K1(σB)]	0.842
	[K1(σS)]	0.793

		Tension/Compression	Bending	Torsion
Notch effect coefficient	[ß]	1.000	1.000	1.000
Geometrical size influence	[K2(d)]	1.000	0.861	0.861
Influence coefficient surface roughness	[KF]	0.907	0.907	0.946
Surface stabilization factor	[KV]	1.000	1.000	1.000
Total influence coefficient	[K]	1.103	1.264	1.218
Present safety for endurance limit:				
Equivalent mean stress (N/mm²)	[σmV]		2.545	
Equivalent mean stress (N/mm²)	[τmV]		1.469	
Fatigue limit of part (N/mm²)	[σWK]	213.708	233.066	145.141

Influence coefficient of mean stress sensitivity.

	[ψσK]	0.222	0.247	0.140
Permissible amplitude (N/mm²)	[σADK]	34.698	225.599	127.339
Safety against fatigue	[S]		10.877	
Required safety against fatigue	[Smin]		1.200	
Result (%)	[S/Smin]		906.4	

Present safety
for proof against exceed of yield point:

Static notch sensitivity factor	[K2F]	1.000	1.200	1.200
Increase coefficient	[γF]	1.000	1.000	1.000
Yield stress of part (N/mm²)	[σFK]	388.575	466.290	269.213
Safety yield stress	[S]		14.396	
Required safety	[Smin]		1.200	
Result (%)	[S/Smin]		1199.6	

Remarks:

- The shearing force is not considered in the analysis specified in DIN 743.
- Cross section with interference fit:
  The notching factor for the light fit case is no longer defined in DIN 743.
  The values are imported from the FKM-Guideline..

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